Supplementary material

Unravelling the structure-performance relationship over iron-based Fischer-Tropsch synthesis by depositing the iron carbonyl in syngas on SiO_2 in a fixed-bed reactor

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Supplementary Figures and Tables

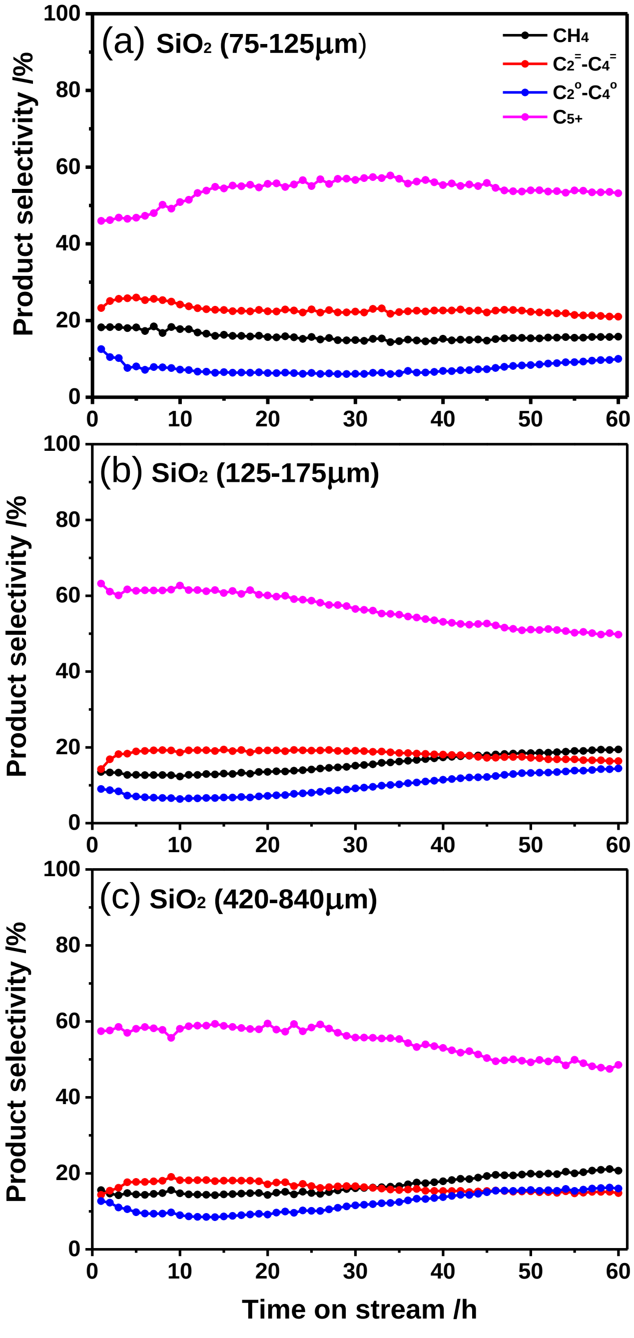

Fig. S1 Effect of the SiO_2 particle size on the product selectivity with TOS over *in situ* deposited Fe catalysts during the FTS reaction as listed in Table 1.

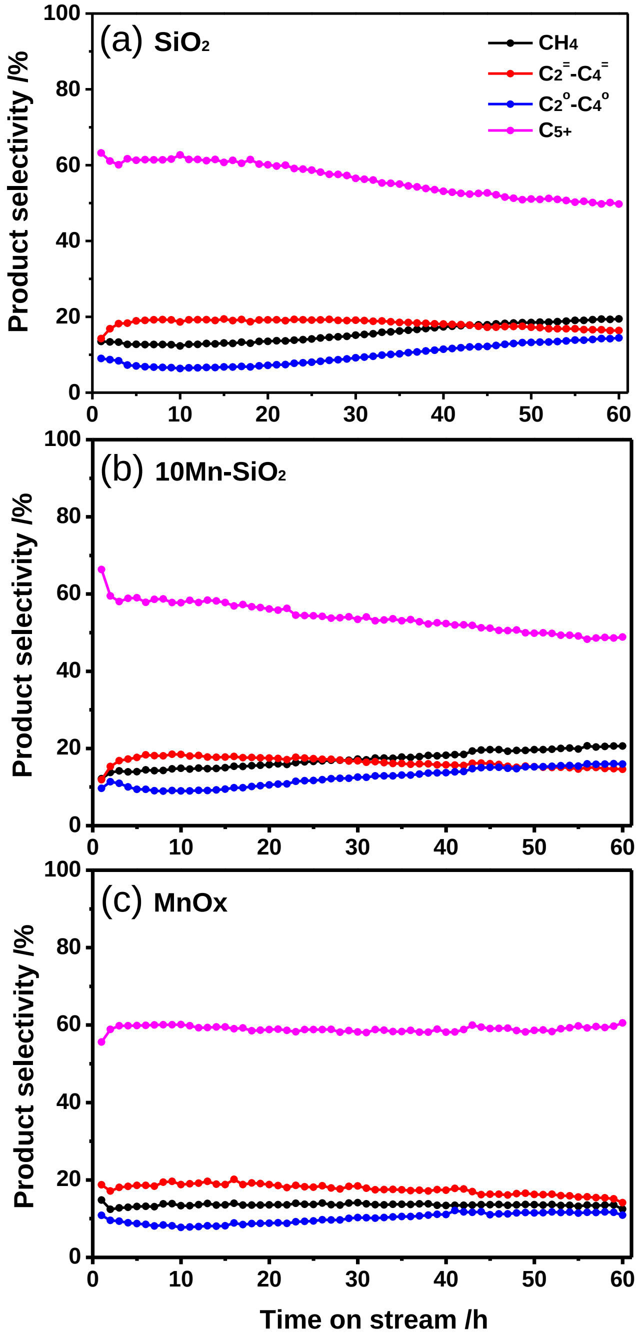

Fig. S2 Effect of the support character on the product selectivity with TOS over *in situ* deposited Fe catalysts during the FTS reaction as listed in Table 2.

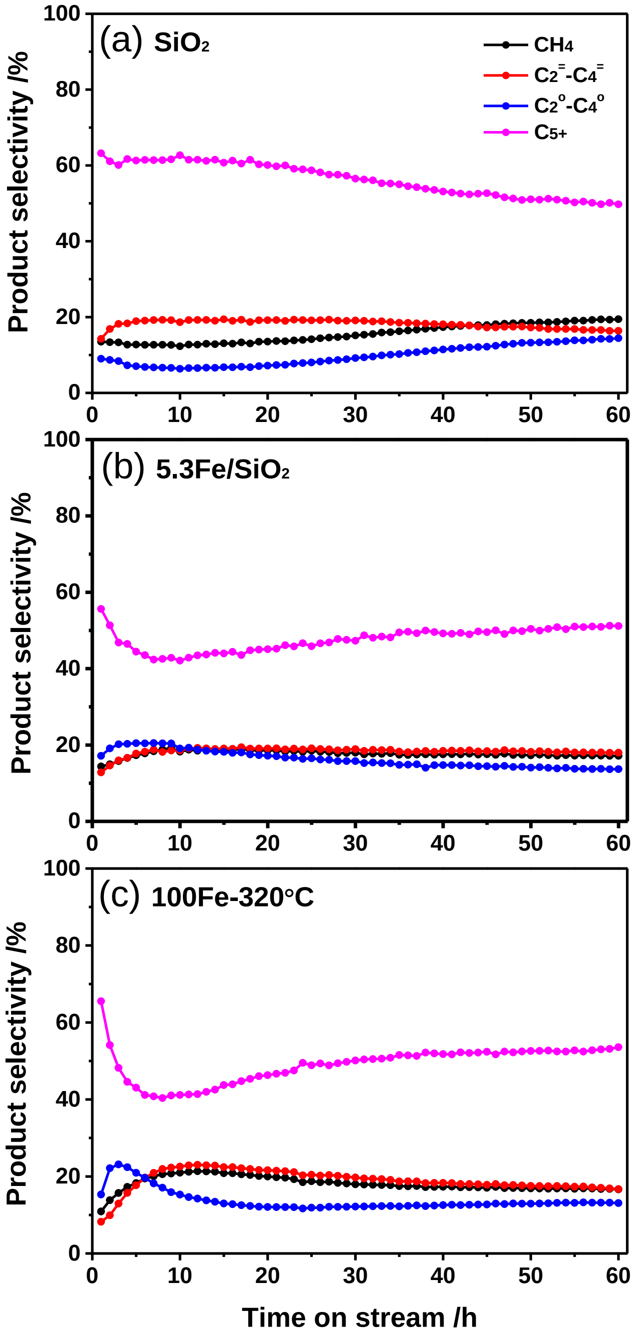

Fig. S3 Effect of the catalyst preparation method on the product selectivity with TOS over as-prepared catalysts as listed in Table 3.

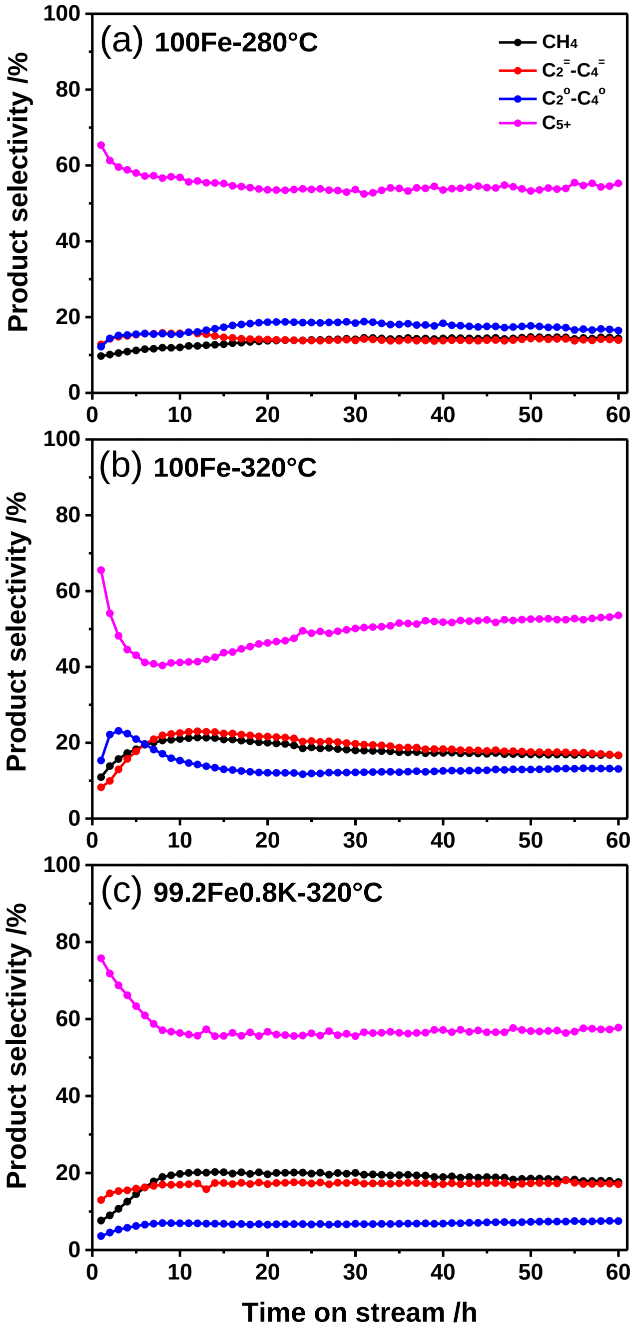

Fig. S4 Effect of the reaction temperature and K promoter on the product selectivity with TOS over as-prepared catalysts as listed in Table 4.

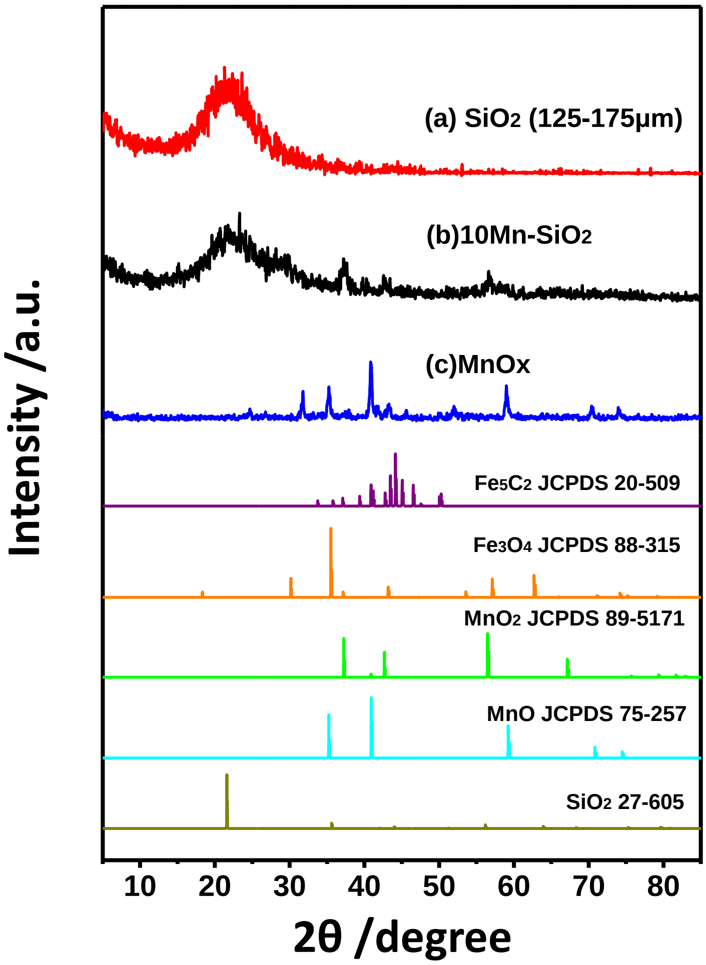

Fig. S5. XRD pattern of *in situ* deposited spent Fe catalysts with different support after reaction of 60 h as the catalytic results are shown in Table 2.

Table S1. BET surface area, pore volume and pore size of the fresh samples.

Sample	Surface area[a] (m^2/g)	Pore volume[b] (cm^3/g)	Pore size[b] (nm)
SiO_2	185.6	1.16	18.9
5.3Fe/SiO_2	189.1	0.93	15.5
100Fe	51.3	0.22	13.1
99.2Fe0.8K	51.7	0.20	13.1

[a]Determined by BET method; [b]Determined by BJH method.

Table S2. Mössbauer spectra of the calcined, reduced and spent 5.3Fe/SiO_2 catalysts[a].

Catalysts	H(KOe)	IS(mm s^{-1})	QS(mm s^{-1})	$\Gamma/2$(mm s^{-1})	Phase ascription	A(%)
Calcined		0.33	0.72	0.28	Fe(III)	**100**
Reduced		0.35	0.75	0.30	Fe(III)	**39.9**
		0.69	0.78	0.65	Fe(II)	**60.1**
Spent	437.97	0.58	-0.01	0.65	Fe_3O_4(A)	**47.0**
	173.8	0.30	-0.12	0.49	χ-Fe_5C_2(I)	**14.5**
		0.37	0.86	0.38	Fe(III)	**24.5**
		0.68	2.14	0.67	Fe(II)	**13.9**

[a] Definitions: H, hyperfine magnetic field; IS, isomer shift (relative to α-Fe); QS, quadrupole splitting; Γ, fwhm; A, relative spectral area.

Table S3. Mössbauer spectra of the spent Fe catalysts after reaction of 60 h as the catalysts were prepared with different methods and evaluated at different reaction conditions (shown in Tables 1- 4) [a].

Catalysts	H(KOe)	IS(mm s^{-1})	QS(mm s^{-1})	Γ/2(mm s^{-1})	Phase ascription	A(%)
SiO_2	220.85	0.42	-0.05	0.21	χ-Fe_5C_2(III)	**30.9**
75-125 μm	182.98	0.36	0.01	0.22	χ-Fe_5C_2(II)	**27.4**
	100.90	0.20	-0.10	0.21	χ-Fe_5C_2 (I)	**19.6**
		0.39	0.26	1.12	Fe(II)	22.0
SiO_2	216.44	0.32	0.07	0.36	χ-Fe_5C_2(III)	**32.7**
125-175 μm	166.94	0.26	-0.07	0.26	χ-Fe_5C_2(II)	**44.7**
	100.12	0.01	-0.08	0.15	χ-Fe_5C_2(I)	**11.1**
		0.11	0.62	0.26	Fe(III)	11.5
SiO_2	218.11	0.33	-0.06	0.31	χ-Fe_5C_2(III)	**33.2**
420-840 μm	168.80	0.20	-0.13	0.18	χ-Fe_5C_2(II)	**38.0**
	101.34	0.18	-0.21	0.25	χ-Fe_5C_2(I)	**15.5**
		0.27	0.87	0.30	Fe(III)	13.3
10Mn-SiO_2	217.81	0.30	-0.04	0.25	χ-Fe_5C_2(III)	**21.3**
	167.75	0.25	-0.10	0.20	χ-Fe_5C_2(II)	**38.3**
	104.18	0.07	-0.40	0.16	χ-Fe_5C_2(I)	**6.1**
		0.37	1.10	0.34	Fe(III)	24.1
		1.01	0.92	0.45	Fe(II)	10.2
5.3Fe/SiO_2	437.97	0.58	-0.01	0.65	Fe_3O_4(B)	47.0
	173.8	0.30	-0.12	0.49	χ-Fe_5C_2(II)	**14.5**
		0.37	0.86	0.38	Fe(III)	24.5
		0.68	2.14	0.67	Fe(II)	13.9
100Fe	490.53	0.31	-0.04	0.16	Fe_3O_4(A)	11.1
280°C	459.12	0.65	0.03	0.21	Fe_3O_4(B)	22.7
	216.06	0.25	-0.11	0.18	χ-Fe_5C_2(III)	**25.7**
	184.95	0.20	-0.02	0.22	χ-Fe_5C_2(II)	**28.8**
	112.34	0.20	-0.05	0.20	χ-Fe_5C_2(I)	**11.6**
100Fe	489.02	0.30	0.03	0.12	Fe_3O_4(A)	3.0
320°C	456.95	0.71	0.29	0.19	Fe_3O_4(B)	5.2
	216.42	0.25	-0.11	0.19	χ-Fe_5C_2(III)	**32.4**
	185.66	0.21	0.01	0.23	χ-Fe_5C_2(II)	**36.2**
	107.60	0.18	0	0.28	χ-Fe_5C_2(I)	**23.2**
99.2Fe0.8K	216.24	0.26	-0.10	0.17	χ-Fe_5C_2(III)	**36.7**
320°C	183.50	0.18	-0.02	0.22	χ-Fe_5C_2(II)	**43.0**
	109.32	0.20	-0.10	0.19	χ-Fe_5C_2(I)	**20.3**

[a] Definitions: H, hyperfine magnetic field; IS, isomer shift (relative to α-Fe); QS, quadrupole splitting; Γ, fwhm; A, relative spectral area.